

Priority Queue Structure

Multi-Level Priority Scheduling

High Priority

Levels 0-3

CPU-intensive
Background tasks
Real-time processes
Low affinity

Medium Priority

Levels 4-8

Normal processes
Most user tasks
Batch operations
Default priority

Low Priority

Levels 9-12

Interactive tasks
UI responsiveness
I/O bound
High affinity

Real-Time EDF

Level 13

Earliest Deadline
First scheduling
Critical tasks
Hard deadlines

Queue Structure

```
Schedq runq[Nrq];  
ulong runvec; // Bitmap of non-empty queues  
• FIFO within each priority level
```

Priority Calculation

```
reprioritize(p) = 13 - (p->cpu * 13 / 1000)  
• Lower CPU usage → Higher priority  
• Priority decays over time
```

Scheduler Selection Logic

1. Search queues from highest priority (13) to lowest (0)
2. Check CPU affinity: Prefer matching CPU, steal if necessary
3. Cooperative scheduling: Check readied process first